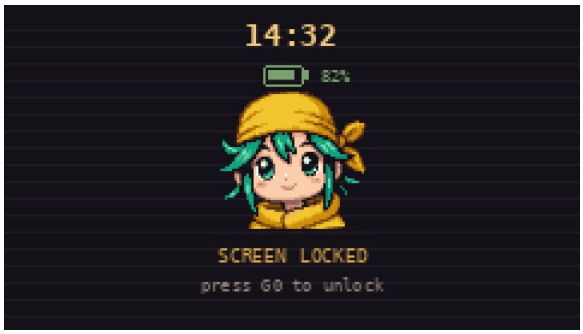
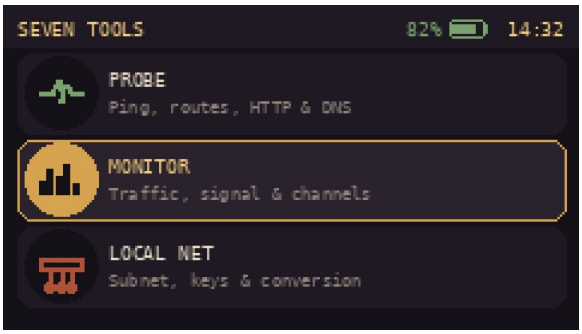
SEVEN TOOLS OF THE BANDIT

Sysadmin Swiss-army knife for the M5Stack Cardputer v1

Powered by Shin-Lima_Kun



Lock screen with Raimu



Main menu

What is it?

Seven Tools of the Bandit is a firmware for the M5Stack Cardputer v1 that packs 29 network, diagnostic and administration tools into one pocket-sized device. Everything is driven from the physical keyboard and shown on the 240×135 px screen, with an earth-tone, old-gold and terminal-green interface.

The tools are organized into 7 categories: WiFi, Probe, Monitor, Local Net, USB HID, Files and Server.

Requirements and important notes

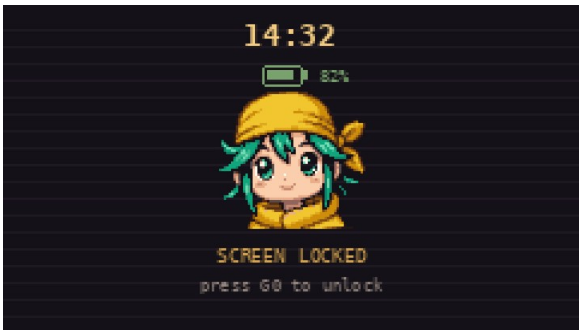
- **Hardware:** M5Stack Cardputer v1 (ESP32-S3) with a microSD card for the file tools, network history and saved networks.
- **USB mode — TinyUSB:** the project needs ARDUINO_USB_MODE=0 in platformio.ini. That's the only mode that lets the device act as a USB keyboard or mouse (HID); it's different from the «Hardware CDC» mode used by other Cardputer projects.
- **microSD pins:** CS=GPIO12, SCK=40, MISO=39, MOSI=14. If your unit differs, adjust the SD_SPI_* constants in the source.
- **«Ping» is not ICMP:** it measures TCP connection time to a port — a good indicator of reachability and latency, but not a strict ping.
- **Packet Monitor:** uses the ESP32's promiscuous mode and only counts packets by type (MGMT, DATA, CTRL); it never stores or shows their contents.
- **Not yet tested on hardware:** the keyboard/mouse USB mode and the monitor and file-explorer modules haven't been tested on a physical Cardputer yet — worth double-checking the first time.
- **Responsible use:** port scanning, traceroute and network monitors should only be used on your own networks or with explicit authorization.

General controls

Tool	What it does
; .	Move the selection (up / down)
ENTER	Choose or activate
`	Back or exit
i	Help for the current screen (every module has its own)
M	Mute or unmute sound
G0	Lock or unlock the screen and keyboard without pausing what's running



Startup screen



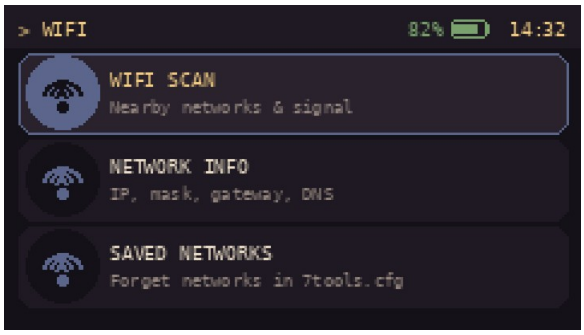
Lock screen with Raimu, clock & battery

The lock screen shows the time (NTP if there's internet, or «+hh:mm» since power-on), the battery level, and Raimu, the project's mascot, with a gentle idle sway. Scans, monitors and the file server keep running in the background while it's locked.

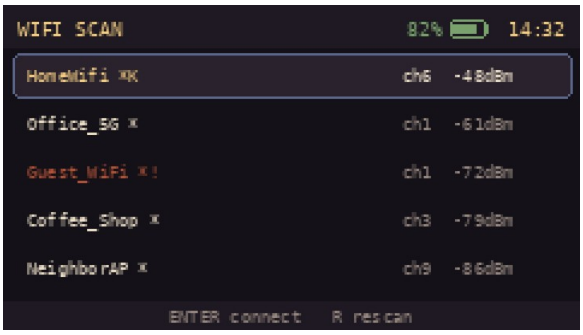
1 · WIFI

Nearby networks, connecting, saved networks, and duplicate-name alerts.

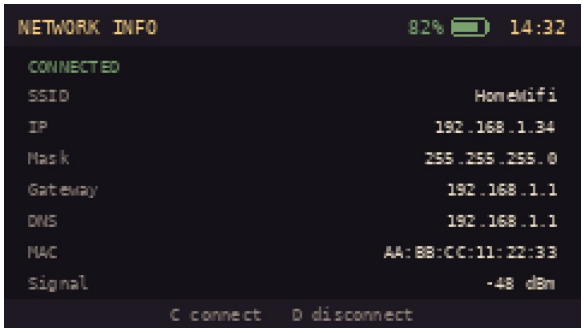
Tool	What it does
WiFi Scan	Lists nearby networks with signal (RSSI) and channel. ENTER connects; K marks a password already saved on the SD, and ! flags an SSID used by two routers.
Network Info	Shows IP, mask, gateway, DNS, MAC and signal. C reconnects to the last network; D disconnects.
Saved Networks	Manages the passwords in 7tools.cfg on the microSD and lets you forget a network (with confirmation). The password is never shown.
Twin Alert	Detects repeated WiFi names with a different router MAC: [!] different encryption (suspicious), [?] a saved network, [i] probably a mesh. Only listens; never creates networks.
Network History	Networks seen in your scans with their last signal and time, saved in /7seen.cfg (up to 40).



WiFi submenu



WiFi Scan

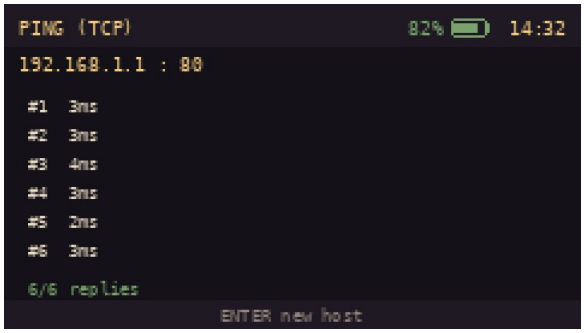


Connected network info

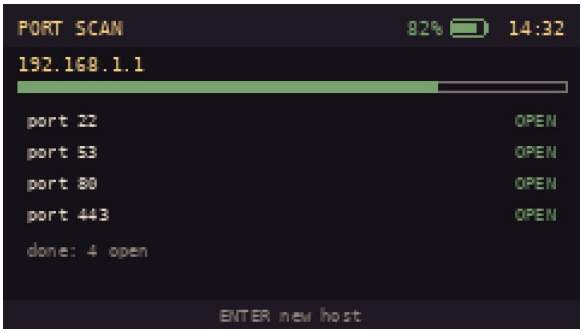
2 · PROBE

Reachability and diagnostic tests against a host, a URL, or a name.

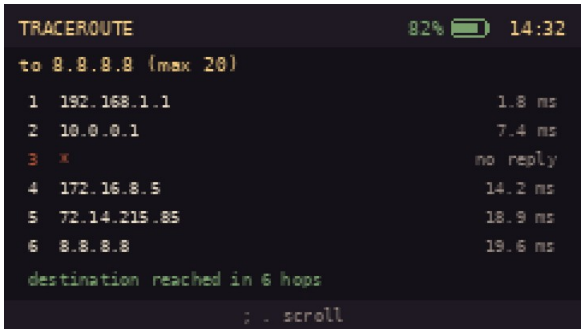
Tool	What it does
Ping (TCP)	Measures TCP connection latency to host:port (80 by default) over 6 attempts.
Port Scan	Tests 16 common TCP ports against a host and flags the open ones.
Traceroute	Shows the hops to a host using ICMP packets with increasing TTL (up to 20). « * » means that hop didn't reply.
HTTP Client	Sends a GET to a URL and shows the status code and headers, without downloading the body. Doesn't validate the certificate over HTTPS.
DNS	Resolves a name to an IP, or looks up the reverse name (PTR) of an IP, using your network's DNS.



Ping TCP



Port Scan

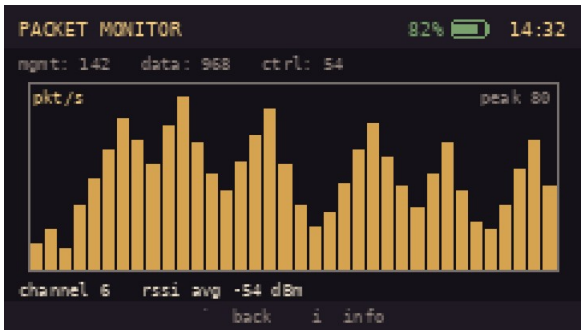


Traceroute

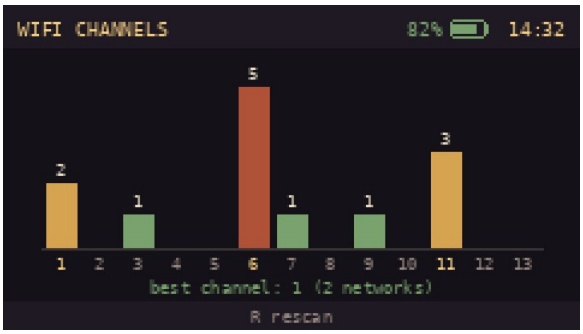
3 · MONITOR

Traffic, signal and WiFi channel occupancy.

Tool	What it does
Packet Monitor	Stacked graph of WiFi traffic per second (MGMT, DATA, CTRL), on a fixed channel or auto-hopping (HOP). Count only; stores no data.
WiFi Channels	Counts networks per channel (1-13) and suggests the best of 1, 6 and 11.
Device Count	Counts how many distinct devices are talking to a chosen network. Doesn't store or show MACs: uses a temporary, randomly-salted hash.
Live Signal	Real-time graph of a network's signal; walk around and watch where it rises. Optional tone (higher pitch = stronger).
Free Channel	Measures each channel's actual traffic combined with neighboring networks to suggest the best channel for your own AP.



Packet Monitor

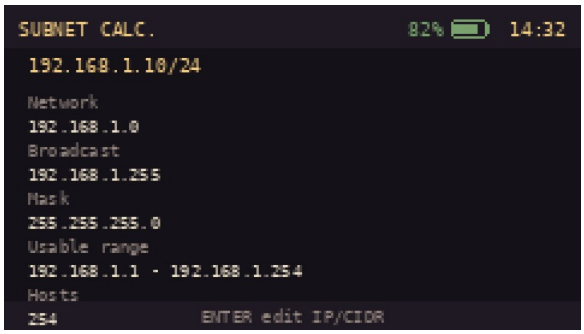


WiFi Channels

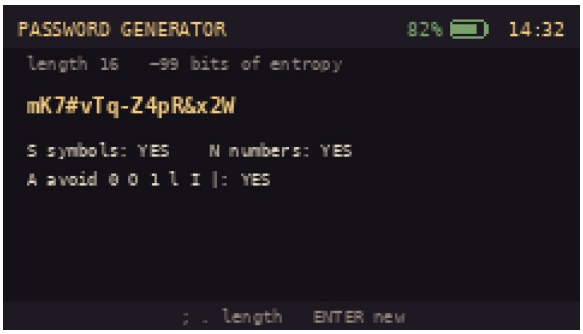
4 · LOCAL NET

Address calculation, validation and conversion, plus network utilities.

Tool	What it does
Subnet Calc.	Given an IP/CIDR, calculates network, broadcast, mask, usable range and host count. Doesn't need WiFi.
Wake on LAN	Sends a «magic packet» to power on a remote machine from its MAC.
Password Generator	Generates random passwords with the chip's RNG, with length, symbols, numbers and «no confusing characters» to taste. Shows the bits of entropy and saves nothing.
Range Check	Checks whether an IP falls inside a subnet (CIDR or with a mask) and flags the network address or the broadcast.
IP Converter	Converts an IP between decimal, binary, hex and integer, and shows its type (private, public, loopback...).



Subnet Calculator

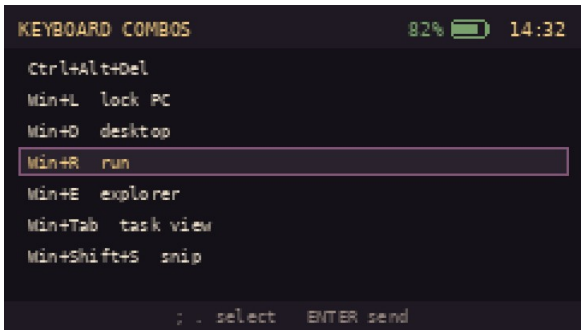


Password Generator

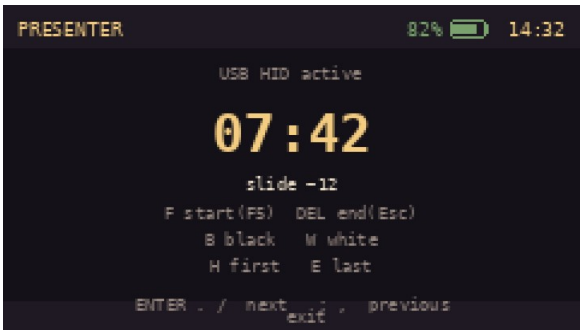
5 · USB HID

The Cardputer connects over USB to a PC and acts as a keyboard, mouse, or remote control.

Tool	What it does
USB Keyboard	What you type is sent to the PC as a real keyboard (ENTER and DEL included). Exit by pressing ` three times in a row.
USB Mouse	The ; . , / keys move the cursor, faster if held. ENTER is left click, DEL is right click.
Keyboard Combos	Sends shortcuts (Ctrl+Alt+Del, Win+L, Alt+Tab, Ctrl+C/V…) with a single ENTER.
Presenter	Remote slide control: next/previous, start (F5), end (Esc), black or white screen, and a timer.



Keyboard Combos

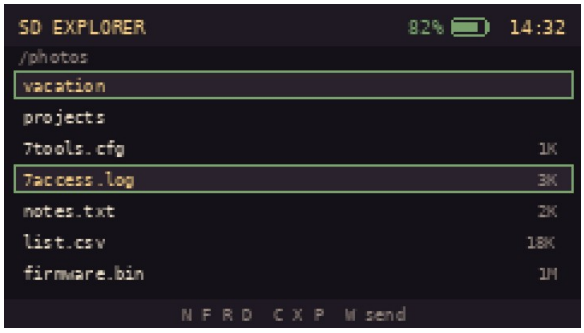


Presenter

6 · FILES

Manages the microSD directly from the device.

Tool	What it does
SD Explorer	Browse folders and files with a text preview. N new folder, F new file, R rename, D delete (confirms), C copy, X cut, P paste, W send over WiFi.
SD Space	Shows card type, capacity, and how much space is used and free.



SD Explorer

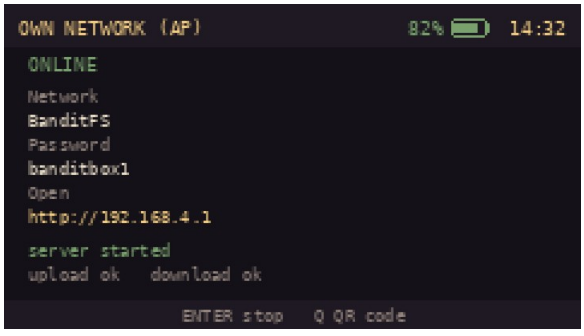


SD Space

7 · SERVER

Serves the microSD as a website to upload and download files from a phone or PC.

Tool	What it does
Own Network (AP)	Creates its own password-protected WiFi and serves the SD. Supports uploading, downloading and deleting files from the web. Q shows a QR code to join and open the site.
On My Network (STA)	Serves the SD on the WiFi you're already connected to. Supports uploading and downloading; web-based deleting is disabled in this mode.
Access Log	Viewer for /7access.log: every upload, download or delete is logged with time, IP and path.



WiFi file server

Recommendation: change the default AP name and password (the fsSSID and fsPass variables in the source) before publishing or using the firmware in a shared environment.

Mascot: Raimu

Raimu, the lock-screen mascot, is an original chibi character designed by Shin-Lima_Kun: yellow headscarf, teal-green hair, big eyes. It's drawn as a 61×57 px sprite with 17 colors, embedded directly in the code with no image files needed.

Note: the screenshots in this document are representations of the interface generated from the firmware's drawing code (real resolution 240×135 px), with example data.